

Project Guide: Intelligent Recovery of Critical Minerals from EV Battery Waste

Computational Modeling, Dashboard, LLM Integration, and Digital Twin Framework

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1 Objective

The objective of this project is to develop an intelligent computational platform for evaluating recovery of critical minerals from end-of-life EV batteries. The platform will integrate:

- Material composition modeling
- Recovery process simulation
- Techno-economic analysis
- Interactive dashboard
- (Advanced) LLM-based decision support
- (Advanced) Digital twin framework

2 Scope

Focus is restricted to EV battery waste:

- Lithium-ion batteries: NMC, LFP, LCO
- Black mass after preprocessing

3 Overall Architecture

User Input → Models → Optimization → Dashboard
(Advanced: LLM Layer + Digital Twin Integration)

4 Core Methodology (Mandatory)

4.1 Step 1: Battery Composition Model

Develop a database of battery compositions.

Task:

- Create Python dictionaries for NMC, LFP, LCO
- Input: battery type and mass

4.2 Step 2: Recovery Estimation

$$M_i = x_i \cdot M_{\text{total}} \cdot \eta_i \quad (1)$$

Task:

- Assign recovery efficiencies
- Compute recovered metal quantities

4.3 Step 3: Process Selection

Rule-based approach:

- Hydrometallurgy $\rightarrow$ high selectivity
- Pyrometallurgy $\rightarrow$ mixed feedstock

4.4 Step 4: Techno-Economic Model

Revenue:

$$R = \sum_i M_i P_i \quad (2)$$

Profit:

$$\Pi = R - C \quad (3)$$

4.5 Step 5: Dashboard Development

Use Streamlit.

Components:

- Input panel
- Output (materials recovered)
- Process recommendation
- Cost vs revenue

5 Advanced Module 1: LLM Integration

5.1 Objective

Introduce a Large Language Model (LLM) as an intelligent interface for interacting with the computational tool.

5.2 Role of LLM

The LLM should NOT replace models. It should:

- Interpret user queries
- Call backend functions
- Explain results in natural language

5.3 Example Queries

- “What is the best recovery route for 500 kg NMC waste?”
- “How to maximize cobalt recovery?”
- “What happens if lithium price decreases by 20%?”

5.4 Implementation Strategy

- Use API-based LLM
- Define backend functions:
 - `compute_recovery()`
 - `cost_analysis()`
 - `process_selection()`
- Use structured JSON outputs

5.5 Dashboard Integration

Add a chat panel:

- User input (text)
- LLM response
- Linked to computed outputs

6 Advanced Module 2: Digital Twin Framework

6.1 Objective

Develop a simplified digital twin of an EV battery recycling process.

6.2 Definition

A digital twin is a virtual representation of a physical system that updates dynamically using data.

6.3 System Components

- Input stream: battery waste
- Process units:
 - Leaching reactor
 - Separation unit
 - Metal recovery unit
- Output stream: recovered metals

6.4 Mathematical Representation

Mass balance:

$$\frac{dM_i}{dt} = \text{Input} - \text{Output} + \text{Reaction} \quad (4)$$

6.5 Simulation Tasks

- Model time-dependent recovery
- Introduce efficiency variation
- Simulate process performance

6.6 Integration with Dashboard

- Real-time updating plots
- Sliders for:
 - Temperature
 - Recovery efficiency
 - Feed rate

7 Validation

- Compare with literature
- Sensitivity analysis
- Scenario testing

8 Deliverables

- Python modules (models + economics)
- Streamlit dashboard
- LLM interface (basic)
- Digital twin simulation (simplified)
- Final report

9 Project Timeline (Suggested)

- Weeks 1–2: Composition + recovery model
- Weeks 3–4: Cost model + dashboard
- Weeks 5–6: LLM integration
- Weeks 7–8: Digital twin prototype

10 Future Extensions

- Physics-based leaching models
- AI optimization
- Industrial deployment

11 Conclusion

This project builds a next-generation intelligent platform for EV battery recycling by combining materials science, process engineering, AI, and digital twin concepts. It has strong potential for research, commercialization, and industrial application.